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CAPE SPENCER, AK, USA

WMO: **997804**

Lat: **58.1989N** Lon: **136.6400W** Elev: **85** StdP: **14.65** Time Zone: **-9.00 (NAK)** Period: **86-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	17.0	20.9	0.2	5.6	20.2	5.1	7.2	25.8	58.8	24.5	52.9	23.8	37.4	70	0.923	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	7.0	61.1	52.9	58.4	53.5	57.0	53.5	56.6	58.5	55.5	57.2	54.6	56.3	12.2	80	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	56.1	67.2	57.0	54.9	64.3	56.0	53.8	61.7	55.2	24.1	58.4	23.6	57.3	23.0	56.7	61.3	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	47.0	41.7	36.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(4)
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	43.2	35.7	33.7	35.9	40.2	46.0	50.0	52.5	52.9	50.8	45.0	39.2	36.2	(6)
(7)			DBStd	7.99	6.07	5.49	5.31	3.21	3.20	2.96	2.29	2.37	2.74	3.07	4.59	5.18	(7)
(8)			HDD50	2721	443	455	437	295	130	31	2	1	20	156	325	426	(8)
(9)			HDD65	7940	908	875	902	744	588	450	387	376	425	619	775	891	(9)
(10)			CDD50	257	0	0	0	0	7	32	80	91	45	2	0	0	(10)
(11)			CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)			CDH74	9	0	0	0	0	0	2	5	2	0	0	0	0	(12)
(13)			CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind		WSAvg	17.5	23.9	22.0	20.8	16.2	13.0	12.1	10.8	11.7	15.0	18.9	22.9	22.8	(14)
(15)	Precipitation		PrecAvg	71.4	7.4	5.8	5.2	3.6	3.0	2.4	3.3	4.6	8.5	10.2	8.7	8.7	(15)
(16)			PrecMax	95.4	13.2	9.3	11.7	6.7	6.1	4.4	6.1	8.6	12.0	16.2	16.1	17.1	(16)
(17)			PrecMin	58.0	3.3	2.0	0.8	1.0	0.7	0.9	1.0	1.7	3.0	4.7	2.8	4.5	(17)
(18)			PrecStd	8.9	2.4	2.1	2.2	1.4	1.4	0.8	1.3	1.7	1.9	2.6	2.8	3.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	43.9	49.5	51.0	61.0	68.4	63.6	68.0	62.3	53.5	48.8	46.1	(19)	
(20)			MCWB	45.5	41.0	44.5	43.5	47.0	54.4	53.4	54.6	51.0	50.8	46.3	44.3	(20)	
(21)		2%	DB	46.5	42.5	45.1	47.5	55.2	58.5	59.2	60.3	58.3	51.7	47.3	44.9	(21)	
(22)			MCWB	44.0	40.2	42.4	42.3	45.8	52.7	56.0	53.5	51.3	49.5	44.8	42.9	(22)	
(23)		5%	DB	44.0	41.2	42.9	45.9	52.3	56.0	57.5	58.1	56.3	50.4	45.6	43.9	(23)	
(24)			MCWB	41.9	39.2	40.1	41.5	47.1	52.3	55.1	53.8	52.3	47.9	43.0	41.7	(24)	
(25)		10%	DB	42.2	39.8	41.3	44.6	50.1	54.3	56.3	56.7	55.1	49.1	44.5	42.7	(25)	
(26)			MCWB	40.3	37.5	38.3	40.8	46.5	51.4	54.2	53.8	52.1	46.3	42.2	40.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	42.2	45.7	45.4	51.7	56.2	58.4	58.3	56.1	52.0	47.6	44.9	(27)	
(28)			MCDB	48.6	42.7	48.3	49.9	56.1	62.2	59.5	60.6	57.2	53.0	48.1	45.2	(28)	
(29)		2%	WB	44.8	40.4	43.8	43.9	49.9	54.7	57.0	56.4	54.7	50.3	45.8	43.8	(29)	
(30)			MCDB	46.5	41.2	45.6	46.9	53.0	59.0	58.1	58.5	55.9	51.5	46.8	44.6	(30)	
(31)		5%	WB	43.1	38.5	41.9	42.5	48.5	53.4	55.8	55.3	53.5	48.3	43.8	42.3	(31)	
(32)			MCDB	44.4	39.5	44.3	45.1	51.3	56.2	57.1	56.8	55.0	49.9	45.1	43.4	(32)	
(33)		10%	WB	41.1	37.2	39.4	41.7	47.7	52.2	54.8	54.3	52.6	47.0	42.6	41.0	(33)	
(34)			MCDB	42.0	38.3	40.5	44.2	50.3	54.5	56.1	55.9	54.3	48.7	44.0	42.4	(34)	
(35)	Mean Daily Temperature Range		MDBR	4.7	4.9	4.9	5.9	6.3	6.7	6.8	7.0	6.1	4.4	4.5	4.2	(35)	
(36)		5% DB	MCDBR	4.9	6.2	6.5	7.7	10.0	10.4	9.4	9.8	8.6	5.3	5.5	4.4	(36)	
(37)			MCWBR	4.1	5.6	4.5	5.0	5.5	6.0	6.9	6.8	5.8	4.9	5.1	4.2	(37)	
(38)		5% WB	MCDBR	4.9	6.2	6.3	7.2	7.7	8.9	8.5	8.4	7.0	5.1	5.2	3.9	(38)	
(39)			MCWBR	4.6	5.7	4.5	5.0	5.2	5.8	6.8	6.7	5.7	5.4	5.2	4.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.298	0.298	0.323	0.376	0.397	0.383	0.379	0.365	0.345	0.323	0.292	0.261	(40)	
(41)		taud		2.165	2.385	2.343	2.217	2.236	2.355	2.420	2.474	2.499	2.427	2.225	2.136	(41)	
(42)		Ebn at Noon		165	238	262	262	265	271	270	266	254	224	168	141	(42)	
(43)		Edh at Noon		16	21	28	38	40	36	33	29	24	20	15	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg		127	369	772	1241	1595	1690	1476	1154	757	403	169	81	(44)	
(45)		RadStd		16	52	80	99	143	111	107	134	112	53	23	10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	N/A	+2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page